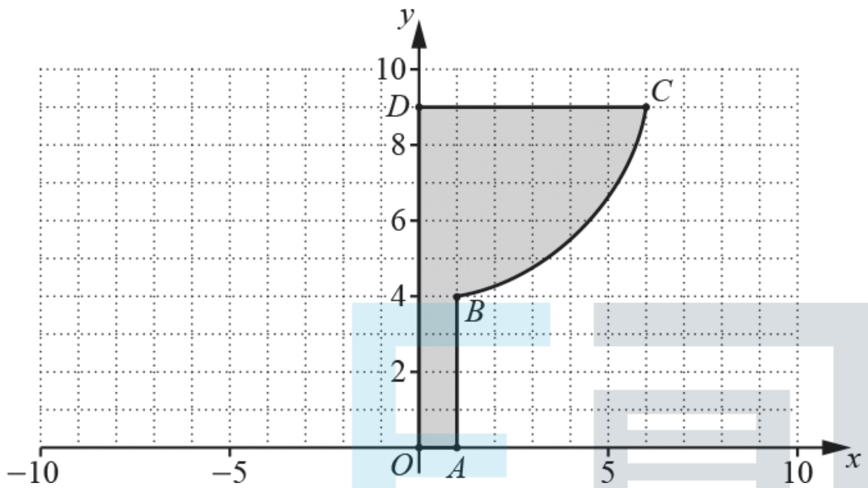
OCR Further Mathematics
Pure Core - Further Calculus
Question Paper
AS and A Level
Further Mathematics A - H235, H245

The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

1 O is the origin of a coordinate system whose units are cm.
The points A, B, C and D have coordinates (1, 0), (1, 4), (6, 9) and (0, 9) respectively.
The arc BC is part of the curve with equation $x^2 + (y - 10)^2 = 37$.
The closed shape $OABCD$ is formed, in turn, from the line segments OA and AB, the arc BC and the line segments CD and DO (see diagram).
A funnel can be modelled by rotating $OABCD$ by 2π radians about the y-axis.



Find the volume of the funnel according to the model. [3]

2(a) You are given that $f(x) = \tan^{-1}(1 + x)$.

(i) Find the value of f(0).

[1]

(ii) Determine the value of $f'(0)$.

[2]

(iii) Show that $f''(0) = -\frac{1}{2}$

[3]

(b) Hence find the Maclaurin series for f(x) up to and including the term in x^2. [2]

3(a) In this question you must show detailed reasoning.

By using an appropriate Maclaurin series prove that if $x > 0$ then $e^x > 1 + x$. [2]

(b) Hence, by using a suitable substitution, deduce that $e^t > et$ for $t > 1$. [1]

(c) Using the inequality in part above, and by making a suitable choice for t, determine which is greater, e^π or π^e.

[4]

4(a) Use differentiation to find the first two non-zero terms of the Maclaurin expansion of $\ln\left(\frac{1}{2}+\cos x\right)$

[3]

(b) By considering the root of the equation $\ln\left(\frac{1}{2}+\cos x\right)=0$ deduce that $\pi \approx 3\sqrt{3\ln\left(\frac{3}{2}\right)}$

[5]

5 Show that $\int_5^{\infty}(x-1)^{-\frac{3}{2}}\mathrm{d}x = 1$.

6(a) Let $f(x) = \sin^{-1}(x)$.

(i) Determine $f''(x)$. [2]

(ii) Determine the first two non-zero terms of the Maclaurin expansion for $f(x)$. [3]

(iii) By considering the first two non-zero terms of the Maclaurin expansion for f(x), find an approximation to $\int_0^{\frac{1}{2}} \mathrm{f}(x)\,\mathrm{d}x$. Give your answer correct to 6 decimal places. [2]

(b)

By writing f(x) as $\sin^{-1}(x) \times 1$, determine the value of $\int_0^{\frac{1}{2}} \mathrm{f}(x)\,\mathrm{d}x$. Give your answer in exact form. [3]

7

Show that $\int_0^{\frac{1}{\sqrt{3}}} \frac{4}{1-x^4}\,\mathrm{d}x = \ln\left(a+\sqrt{b}\right)+\frac{\pi}{c}$ where *a*, *b* and *c* are integers to be determined. [6]

Exam Papers Practice

8(a) Using exponentials, show that $\cosh 2u \equiv 2\sinh^2 u + 1$. [2]

(b) By differentiating both sides of the identity in part (a) with respect to u, show that $\sinh 2u \equiv 2\sinh u \cosh u$. [1]

(c) Use the substitution $x = \sinh^2 u$ to find $\int \sqrt{\frac{x}{x+1}}\,dx$. Give your answer in the form $a\sinh^{-1} b\sqrt{x} + \mathrm{f}(x)$ where a and b are integers and $\mathrm{f}(x)$ is a function to be determined.

[5]

(d)

Hence determine the exact area of the region between the curve $y = \sqrt{\frac{x}{x+1}}$, the x-axis, the line x - 1 and the line x – 2. Give your answer in the form $p + q \ln r$ where p, q and r are numbers to be determined. [2]

9 Find the mean value of $f(x) = x^2 + 6x$ over the interval [0, 3]. [2]

10(a)

The function sech x is defined by $\operatorname{sech} x = \dfrac{1}{\cosh x}$.

Show that $\operatorname{sech} x = \dfrac{2e^x}{e^{2x} + 1}$ [2]

(b) Using a suitable substitution, find $\int \operatorname{sech} x \, dx$. [4]

11(a) You are given that $y = \tan^{-1}\sqrt{2x}$.

[2]

Find . $\frac{dy}{dx}$

(b)

Show that $\int_{\frac{1}{6}}^{\frac{1}{2}} \frac{\sqrt{x}}{(x+2x^2)} \mathrm{d}x = k\pi$ where k is a number to be determined in exact form. [4]

12 In this question you must show detailed reasoning.

Find $\int_{-1}^{11} \frac{1}{\sqrt{x^2+6x+13}} \mathrm{d}x$ giving your answer in the form ln $\left(p+q\sqrt{2}\right)$ where p and q are integers to be determined.

[7]

13 You are given that $f(x) = \ln(2+x)$.

(a) Determine the exact value of $f'(0)$. [2]

(b) Show that $f''(0) = -\frac{1}{4}$. [2]

(c) Hence write down the first three terms of the Maclaurin series for $f(x)$. [3]

14 (a) By using Euler's formula show that $\cosh(iz) = \cos z$. [3]

(b) Hence, find, in logarithmic form, a root of the equation $\cos z = 2$.
[You may assume that $\cos z = 2$ has complex roots.] [2]

15 In this question you must show detailed reasoning.

Solve the equation $2\cosh^2 x + 5\sinh x - 5 = 0$ giving each answer in the form $\ln\left(p + q\sqrt{r}\right)$ where p and q are rational numbers, and r is an integer, whose values are to be determined. [6]

16

(a) Using the Maclaurin series for $\ln(1 + x)$, find the first four terms in the series expansion for $\ln(1 + 3x^2)$. [2]

(b) Find the range of x for which the expansion is valid. [1]

(c) Find the exact value of the series

$$\frac{3^1}{2\times 2^2} - \frac{3^2}{3\times 2^4} + \frac{3^3}{4\times 2^6} - \frac{3^4}{5\times 2^8} + \ldots$$ [3]

17 In this question you must show detailed reasoning.

Find $\int_2^3 \frac{x+1}{x^3-x^2+x-1}\,\mathrm{d}x$, expressing your answer in the form $a\ln b$ where a and b are rational numbers. [6]

18 In this question you must show detailed reasoning.

Find the exact value of each of the following.

(a) $\displaystyle\int_1^4 \frac{1}{x^2 - 2x + 10}\,\mathrm{d}x$ [3]

(b) The mean value of $\dfrac{1}{\sqrt{1 - x^2}}$ in the interval [0, 0.5] [3]

19 A survey is carried out into the length of time for which customers wait for a response on a telephone helpline. A statistician who is analysing the results of the survey starts by modelling the waiting time, x minutes, by an exponential distribution with probability density function (PDF)

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geqslant 0, \\ 0 & x < 0. \end{cases}$$

(a) In this question you must show detailed reasoning.

The mean waiting time is found to be 5.0 minutes. Show that $\lambda = 0.2$. [4]

(b) Use the model to calculate the probability that a customer has to wait longer than 20 minutes for a response. [2]

In practice it is found that no customer waits for more than 15 minutes for a response. The statistician constructs an improved model to incorporate this fact.

(c) On the diagram below, sketch the following, labelling the curves clearly:

(i) the PDF of the model using the exponential distribution, [1]

(ii) a possible PDF for the improved model. [2]

20

The region R between the x-axis, the curve $y=\frac{1}{\sqrt{p+x^2}}$ and the lines $x=\sqrt{p}$ and $x=\sqrt{3p}$, where p is a positive parameter, is rotated by 2π radians about the x-axis to form a solid of revolution S.

(a) Find and simplify an algebraic expression, in terms of p, for the exact volume of S. [5]

Exam Papers Practice

(b) Given that R must lie entirely between the lines $x = 1$ and $x = \sqrt{48}$ find in exact form

- the greatest possible value of the volume of S
- the least possible value of the volume of S. [3]

21 In this question you must show detailed reasoning.

(a) Find $\int_{\frac{1}{4}\pi}^{\frac{1}{3}\pi} 2\tan x \, dx$ giving your answer in the form $\ln p$. [3]

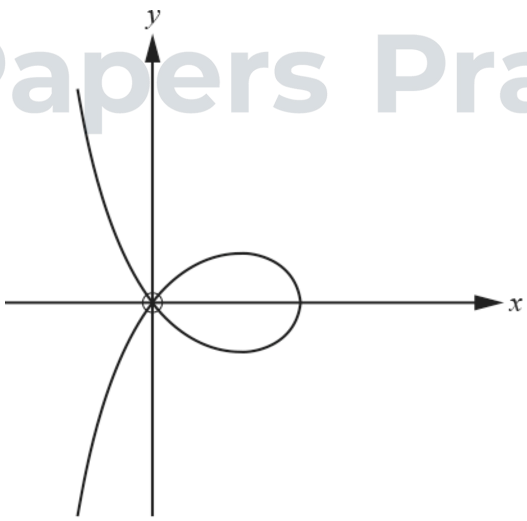
(b) Show that $\int_{0}^{\frac{1}{2}\pi} 2\tan x \, dx$ undefined explaining your reasoning. [2]

22 (a) (i) A curve has polar equation $r = 2 - \sec\theta$. Show that the cartesian equation of the curve can be written in the form

$$y^2 = \left(\frac{2x}{x+1}\right)^2 - x^2.$$

[4]

The figure shows a sketch of part of the curve with equation $y^2 = \left(\frac{2x}{x+1}\right)^2 - x^2$.



(ii) Explain why the curve is symmetrical in the x-axis. [1]

(iii) The line $x = a$ is an asymptote of the curve. State the value of a. [1]

(b) The enclosed loop shown in the figure is rotated through $180°$ about the x-axis.

Find the exact volume of the solid formed. [8]

23 In an experiment, at time t minutes there is Q grams of substance present.

It is known that the substance decays at a rate that is proportional to $1 + Q^2$. Initially there are 100 grams of the substance present and after 100 minutes there are 50 grams present.

Find the amount of the substance present after 400 minutes. [8]

24 You are given that $f(x) = (1 - a \sin x)e^{bx}$ where a and b are positive constants. The first three terms in the Maclaurin expansion of $f(x)$ are $1 + 2x + \frac{3}{2}x^2$.

(a) Find the value of a and the value of b. [6]

(b) Explain if there is any restriction on the value of x in order for the expansion to be valid. [1]

25 By using a suitable substitution, which should be stated, show that

$$\int_{\frac{3}{2}}^{\frac{5}{2}} \frac{1}{\sqrt{4x^2-12x+13}}\,dx = \frac{1}{2}\ln\left(1+\sqrt{2}\right)$$ [6]

26 In this question you must show detailed reasoning.

Show that $\displaystyle\int_0^2 \frac{2x^2+3x-1}{x^3-3x^2+4x-12}\,\mathrm{d}x = \frac{3}{8}\pi - \ln 9$. [12]

27 In this question you must show detailed reasoning.

An ant starts from a fixed point O and walks in a straight line for 1.5 s. Its velocity, v cm s^{-1}, can be modelled by $v = \frac{1}{\sqrt{9-t^2}}$.

By finding the mean value of v in $0 \le t \le 1.5$, deduce the average velocity of the ant. [5]

28

It is given that $y = \tan^{-1}\left(\dfrac{x}{x+1}\right)$.

(i) Show that, when $x = 0$, $\dfrac{d^2y}{dx^2} = -2$. [4]

(ii) Find the Maclaurin's series for $\tan^{-1}\left(\dfrac{x}{x+1}\right)$ up to and including the term in x^2. [2]

29 By first completing the square in the denominator, find the exact value of

$$\int_3^4 \frac{1}{x^2 - 6x + 10}\,\mathrm{d}x.$$ [4]

30

(a) Show that $\frac{\mathrm{d}}{\mathrm{d}x}\left(\sinh^{-1}(2x)\right) = \frac{2}{\sqrt{4x^2+1}}$. [2]

(b) Find $\int \frac{1}{\sqrt{2-2x+x^2}}\,\mathrm{d}x$. [3]

31 (a) Use the Maclaurin series for $\sin x$ to work out the series expansion of $\sin x \sin 2x \sin 4x$ up to and including the term in x^3. [4]

(b) Hence find, in exact surd form, an approximation to the least positive root of the equation $2\sin x \sin 2x \sin 4x = x$. [3]

32 In this question you must show detailed reasoning.

Evaluate $\int_0^{\infty} 2xe^{-x}\mathrm{d}x$.

[You may use the result $\lim_{x\to\infty} xe^{-x}=0$.] [4]

33 In this question you must show detailed reasoning.

The finite region R is enclosed by the curve with equation $y=\frac{8}{\sqrt{16+x^2}}$, the x-axis and the lines x = 0 and x = 4. Region R is rotated through 360° about the x-axis. Find the exact value of the volume generated. [4]

34 It is given that $\mathrm{f}(x)=\frac{x(x-1)}{(x+1)(x^2+1)}$. Express f ($x$) in partial fractions and hence find the exact value of $\int_0^1 \mathrm{f}(x)\,\mathrm{d}x$.

[6]

35 It is given that $y = \tan^{-1} 2x$.

(i) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ and show that $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4x\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 0$. [3]

(ii) Find the Maclaurin series for y up to and including the term in x^3. Show all your working. [4]

(iii) The result in part (ii), together with the value $x = \frac{1}{2}$, is used to find an estimate for π. Show that this estimate is only correct to 1 significant figure.

[2]

36 It is given that $f(x) = \ln(1 + \sin x)$. Using standard series, find the Maclaurin series for $f(x)$ up to and including the term in x^3.

[4]

37

By first completing the square, find the exact value of $\int_{\frac{1}{2}}^{1} \frac{1}{\sqrt{2x - x^2}}\,dx$.

[5]

38

(i) Given that $y = \ln(1 + \sin x) - \ln(\cos x)$, show that $\frac{dy}{dx} = \frac{1}{\cos x}$.

[4]

(ii) Using this result, evaluate $\int_{0}^{\frac{1}{3}\pi} \sec x\,dx$, giving your answer as a single logarithm.

[3]

39 It is given that $y = \sin^{-1}2x$.

(i) Using the derivative of $\sin^{-1}x$ given in the List of Formulae (MF1), find $\frac{\mathrm{d}y}{\mathrm{d}x}$.

[1]

(ii) Show that $(1-4x^2)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 4x\frac{\mathrm{d}y}{\mathrm{d}x}$.

[3]

(iii) Hence show that $(1-4x^2)\frac{\mathrm{d}^3y}{\mathrm{d}x^3} - 12x\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} = 0$.

[2]

(iv) Using your results from parts (i), (ii) and (iii), find the Maclaurin series for $\sin^{-1}2x$ up to and including the term in x^3.

[3]

40

(i) Use algebraic division to express $\frac{x^3 - 2x^2 - 4x + 13}{x^2 - x - 6}$ in the form $Ax + B + \frac{Cx + D}{x^2 - x - 6}$, where A, B, C and D are constants.

[4]

(ii) Hence find $\int_4^6 \frac{x^3 - 2x^2 - 4x + 13}{x^2 - x - 6}\,\mathrm{d}x$, giving your answer in the form $a + \ln b$.

[7]

41 You are given that $f(x) = e^{-x} \sin x$.

(i) Find $f(0)$ and $f'(0)$.

[3]

(ii) Show that $f''(x) = -2f'(x) - 2f(x)$ and hence, or otherwise, find $f''(0)$.

[4]

(iii) Find a similar expression for $f'''(x)$ and hence, or otherwise, find $f'''(0)$.

[2]

(iv) Find the Maclaurin series for $f(x)$ up to and including the term in x^3.

[2]

42

Find $\int_0^2 \frac{1}{\sqrt{4+x^2}}\,dx$, giving your answer exactly in logarithmic form.

[3]

43 It is given that $f(x) = \ln(1 + x^2)$.

(i) Using the standard Maclaurin expansion for $\ln(1 + x)$, write down the first four terms in the expansion of $f(x)$, stating the set of values of x for which the expansion is valid.

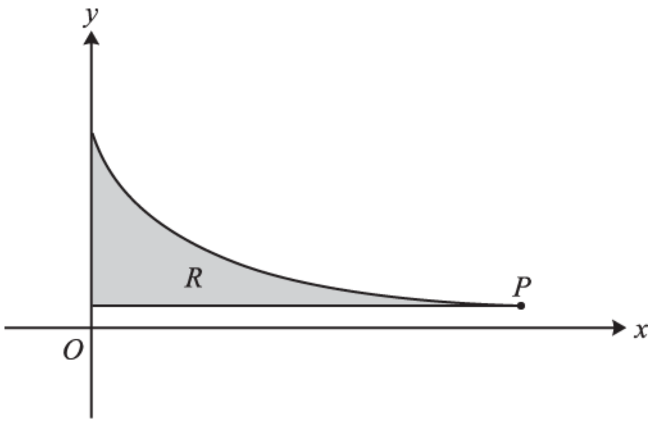
[3]

(ii) Hence find the exact value of

$$1-\frac{1}{2}\left(\frac{1}{2}\right)^2+\frac{1}{3}\left(\frac{1}{2}\right)^4-\frac{1}{4}\left(\frac{1}{2}\right)^6+\dots.$$

[2]

44



The diagram shows the curve $y = \sqrt{\frac{3}{4x+1}}$ for $0 \le x \le 20$. The point P on the curve has coordinates $\left(20, \frac{1}{9}\sqrt{3}\right)$. The shaded region R is enclosed by the curve and the lines $x = 0$ and $y = \frac{1}{9}\sqrt{3}$.

(i) Find the exact area of R. [4]

(ii) Find the exact volume of the solid obtained when R is rotated completely about the x-axis. [6]

45

(i) Given that $y = \cosh^{-1} x$, show that $y = \ln\left(x + \sqrt{x^2 - 1}\right)$. [4]

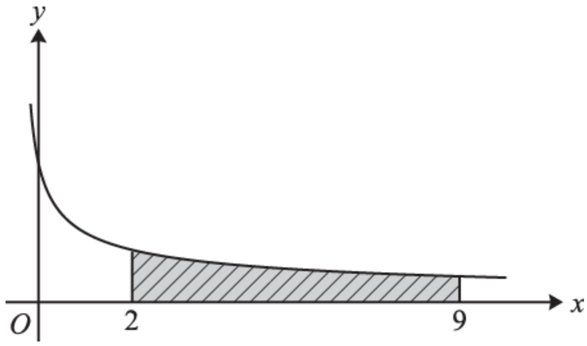
(ii) Show that $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2 - 1}}$. [2]

(iii) Solve the equation $\cosh x = 3$, giving your answers in logarithmic form. [3]

46

By using the substitution $t = \tan \frac{1}{2}\theta$, find $\int_0^{\frac{1}{2}\pi} \frac{1}{1 + \cos\theta} \, d\theta$. [5]

47



The diagram shows the curve $y = \frac{6}{\sqrt{3x + 1}}$. The shaded region is bounded by the curve and the lines $x = 2$, $x = 9$ and $y = 0$.

(i) Show that the area of the shaded region is $4\sqrt{7}$ square units.

[4]

(ii) The shaded region is rotated completely about the x-axis. Show that the volume of the solid produced can be written in the form $k\ln 2$, where the exact value of the constant k is to be determined.

[5]

48

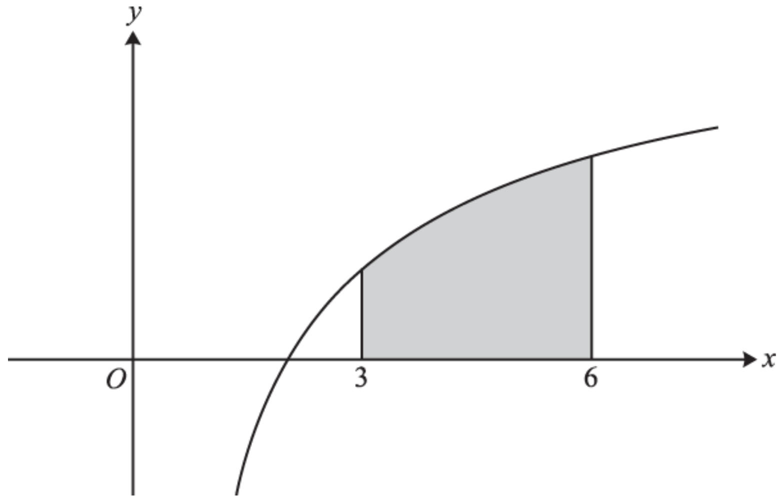
It is given that $f(x) = \tanh^{-1}\left(\frac{1 - x}{3 + x}\right)$ for $x > -1$.

(i) Show that $f''(x) = \frac{1}{2(x + 1)^2}$. [6]

(ii) Hence find the Maclaurin series for f(x) up to and including the term in x^2.

[4]

49



The diagram shows part of the curve $y = \ln(\ln(x))$. The region between the curve and the x-axis for $3 \leqslant x \leqslant 6$ is shaded.

(i) By considering n rectangles of equal width, show that a lower bound, L, for the area of the shaded region is

$$\frac{3}{n}\sum_{r=0}^{n-1}\ln\left(\ln\left(3+\frac{3r}{n}\right)\right).$$

[3]

(ii) By considering another set of n rectangles of equal width, find a similar expression for an upper bound, U, for the area of the shaded region.

[1]

(iii) Find the least value of n for which $U - L < 0.001$.

[4]

END OF QUESTION PAPER